PRATICAL-3

AIM:- Perform the data classification using classification algorithm using R/Python.

STEPS:-

Software required: R 3.5.1

Step 1:- Time series is a series of data points in which each data point is associated with a timestamp. A simple example is the price of a stock in the stock market at different points of time on a given day. Another example is the amount of rainfall in a region at different months of the year. R language uses many functions to create, manipulate and plot the time series data. The data for the time series is stored in an R object called time-series object. It is also a R data object like a vector or data frame.

Step 2:-The time series object is created by using the ts () function.

Syntax

The basic syntax for ts() function in time series analysis is – timeseries.object.name <- ts(data, start, end, frequency)

Following is the description of the parameters used –

- **data is a vector or matrix containing the values used in the time series.**
- **start specifies the start time for the first observation in time series.**
- **end specifies the end time for the last observation in time series.**
- **end specifies the end time for the last observation in time series.**

Except the parameter "data" all other parameters are optional

Consider the annual rainfall details at a place starting from January 2012. We create an R time series object for a period of 12 months and plot it

Step 3:-Code

```
# Get the data points in form of a R vector.

rainfall <-

c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)

# Convert it to a time series object.

rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)

# Print the timeseries data

print(rainfall.timeseries)

# Give the chart file a name.

png(file = "rainfall.png")
```

 # Plot a graph of the time series.

plot(rainfall.timeseries)

 # Save the file.

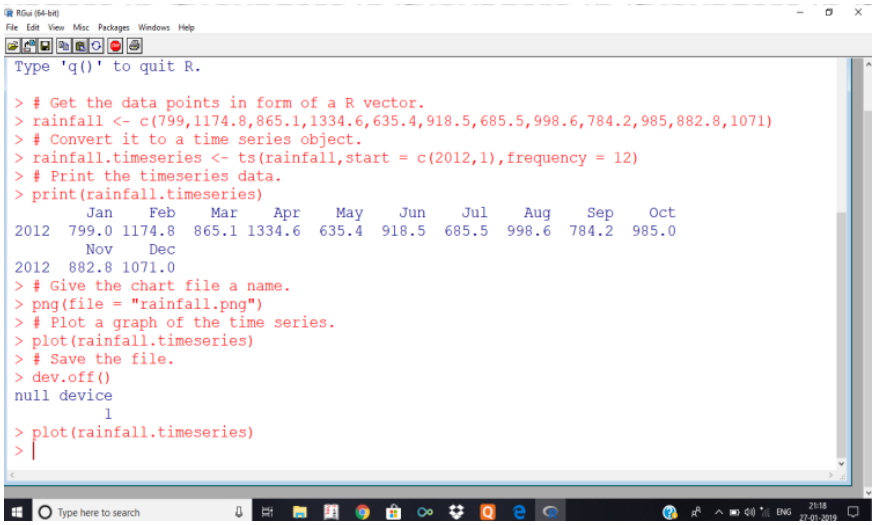
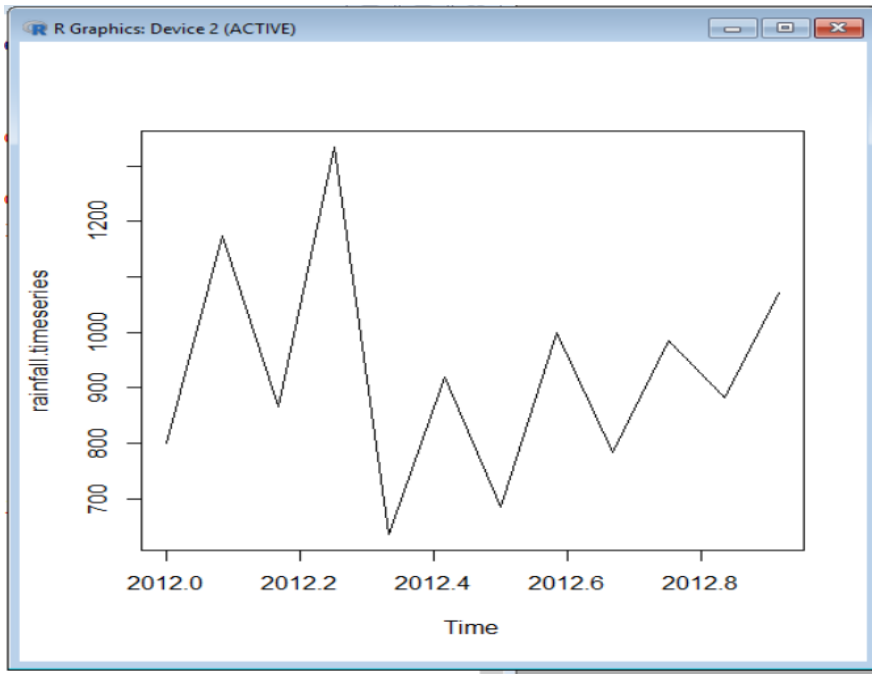
dev.off()

After this again plot to get chart

plot(rainfall.timeseries)

Output:

Jan Feb Mar Apr May Jun Jul Aug Sep 2012 799.0 1174.8 865.1 1334.6 635.4 918.5 685.5 998.6 784.2 Oct Nov Dec 2012 985.0 882.8 1071.0



```
Type 'q()' to quit R.

> # Get the data points in form of a R vector.
> rainfall <- c(799,1174.8,865.1,1334.6,635.4,918.5,685.5,998.6,784.2,985,882.8,1071)
> # Convert it to a time series object.
> rainfall.timeseries <- ts(rainfall,start = c(2012,1),frequency = 12)
> # Print the timeseries data.
> print(rainfall.timeseries)
        Jan    Feb    Mar    Apr    May    Jun    Jul    Aug    Sep    Oct
2012  799.0 1174.8  865.1 1334.6  635.4  918.5  685.5  998.6  784.2  985.0
        Nov    Dec
2012  882.8 1071.0
> # Give the chart file a name.
> png(file = "rainfall.png")
> # Plot a graph of the time series.
> plot(rainfall.timeseries)
> # Save the file.
> dev.off()
null device
          1
> plot(rainfall.timeseries)
>
```